

# Sociology 500: Applied Social Science Statistics (Fall 2026)

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Latest Version: September 7, 2026

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Class: Tuesdays and Thursdays, 10:00–11:20 a.m.

Precept: Fridays, 10:00am–12:00pm

This is the first class in Sociology’s two-course graduate statistics sequence. Starting from only basic math, we build up a foundation for linear regression and its application to causal inference. The course is focused on the tools needed to do research and draws examples from across the social sciences. Students can take it both as a first course in linear regression or as a deeper dive into regression than a typical undergraduate sequence.

## 1 The Basics

The first stop for anything you need as a member of the class should be the Canvas site.

### 1.1 Course Goals

This is the first course in the Department of Sociology’s two-course graduate statistics sequence. Starting from only basic math, we build up a foundation for linear regression and its application to causal inference. Students will learn the statistical and computational principles necessary to perform modern, flexible, and creative analysis of quantitative social data. By the end of this semester, you will be on your way toward (1) engaging with modern, quantitative sociological research and (2) having a foundation for lifelong learning about quantitative methods.

Students enter this course with a huge range of goals related to modern, quantitative sociological research: some students seek to read and understand it, some seek to perform analysis using existing methods, and some seek to develop new methods. You do not need to make any hard choices at the outset about what you want to do, but it can be helpful to formulate your goals and communicate them with us.

We have designed the course to help you

- Critically read and reason about quantitative social science using linear regression techniques grounded in probability theory.
- Conduct, interpret, and communicate results from analysis using multiple regression.
- Explain the limitations of observational data for making causal claims and distinguish between identification and estimation in causal inference.
- Understand the logic and assumptions of several modern designs for making causal claims.
- Write clean, reusable, and reliable R code in the tidyverse style.
- Feel empowered working with data.

The second course in the sequence, SOC 504, will be offered in the spring. The overarching goal of the two-course sequence is to move you from being consumers of quantitative research to producers of it. The capstone of the two-course sequence is the replication and extension project. In this project, completed in Sociology 504, you will choose a paper of interest, reproduce the results, and then extend them to make something new. The projects are generally presented at the department's Graduate Research Day in the Spring during a poster session and written up in a paper.

More broadly, the statistics sequence is designed to get you to a point where you can teach yourself new statistical methods by reading the literature. We can't teach you all the statistical techniques that you will need during your career, but we can prepare you to teach yourself.

This course will require a lot of hard work from all of us; however, we have structured the class to provide you the maximal return on every hour of work you put in. As you read through this syllabus you will find numerous avenues for seeking help. If you are willing to put in the time, we are always happy to help. Please don't be shy about telling us where you need support.

## 1.2 Prerequisites

The most important prerequisite is a willingness to work hard on unfamiliar material. Learning statistical methods is like learning a new language, and it will take time and dedication to master its vocabulary, its grammar, and its idioms. However like studying languages, statistics and programming yield to daily practice and consistent effort.

We intentionally have no formal pre-requisites. Beyond high-school level algebra, it is helpful to have some familiarity with calculus (essentially knowing what derivatives and integrals are in principle even if you forget how to do the mechanics) and basic matrix operations (matrix multiplications and inverses). It will also be helpful if you have some experience with computer programming, specifically the R language. If these concepts are unfamiliar, you should review the materials from our Department of Sociology summer methods camp (<http://pusocmethodscamp.org/>).

Even if you have seen some of the materials in class before (e.g., you had an undergraduate class on linear regression), you will likely find a lot to learn here. By rebuilding the foundations of linear regression from scratch, we help to ensure that everyone is on the same page, can more deeply appreciate the intricacies of these methods, and can have a solid foundation for learning more advanced methods. If you are concerned that you may have already covered the material before, come talk to the instructor.

### 1.3 Preview of Course Structure

- Lectures (Tuesday/Thursday, 10:00–11:20 a.m.):  
Lectures cover the core material and are organized into weeks. In each lecture you will be exposed to the main conceptual ideas of the week and have the opportunity to ask questions. Detailed slides are made available afterwards.
- Lab/Precepts (Friday, 10:00 a.m.–noon):  
We will often refer to this preceptor-led period as 'precept' but administratively it is a 'lab.' Regardless, the goal of this time is to learn the coding skills necessary to complete the problem sets. It also provides an opportunity to ask any followup questions about the material.
- Problem Sets (Thursdays at 8:00pm):  
Learning comes through doing and the problem set is an opportunity to practice the material learned in the prior week. These problem sets will likely be the most time-consuming part of the class (self-reports in prior years suggest they take most people 5-10 hours per week) but they are also where the majority of the learning happens.
- Office Hours (Posted on Canvas)  
The instructional staff will hold weekly office hours; times and locations will be posted on Canvas. Office hours with the instructor are primarily for discussing conceptual material while office hours with the preceptors should be used for assistance with the programming and/or problem sets. You can come to preceptor office hours without any specific questions and just sit and work on the problem sets if you like!

## 2 Materials

### 2.1 Computational Tools

The best way, and often the only way, to learn about data analysis and new statistical procedures is by doing. We will therefore make extensive use of a flexible (open-source and free) statistical software program called R and RStudio, and a number of companion packages in the tidyverse style. Problem sets and written computational work will be completed in Quarto, with code and answers rendered together in a PDF. You will learn how to program in this class, if you do not know already.

### 2.2 Readings

This class uses extremely detailed lecture slides (it is not uncommon to have 100+ slides in a week). We encourage you to think of these (and the lectures) as the primary reading in the class (although

we will occasionally assign additional required reading). If you are someone who benefits a lot from reading material and you find the slides aren't working for you, come talk to me and I will help find a reading that is well situated to your background and interests.

Sometimes though it is just helpful to have a reference book. We include a few by topic below.

- Programming
  - Wickham, Çetinkaya-Rundel, and Golemund. 2023. *R for Data Science, 2nd edition* (available [online for everyone](#))
  - Healy, Kieran. 2018. *Data Visualization: A Practical Introduction*. Princeton University Press. (available [online for everyone](#))
- Probability and Random Variables
  - Blitzstein and Hwang. 2019. *Introduction to Probability, 2nd Edition* ([freely available online](#))
- Regression
  - Angrist, Joshua D. and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton University Press. (available online through [JSTOR](#))
  - Aronow and Miller. 2019. *Foundations of Agnostic Statistics*. Cambridge University Press (available online through the library)
  - Fox, John. 2016 *Applied Regression Analysis and Generalized Linear Models. 3rd Edition*.
  - Shalizi, Cosma Rohilla. *Advanced Data Analysis from an Elementary Point of View*. Draft in progress (freely available online).
- Causal Inference
  - Hernán, Miguel A. and James M. Robins. 2020. *Causal Inference: What If*. Boca Raton: Chapman & Hall/CRC (freely available online).
  - Morgan, Stephen L., and Christopher Winship. 2015. *Counterfactuals and Causal Inference: Methods and Principles for Social Research, 2nd Edition*. Cambridge University Press. (available online through the library)

### 3 Assignments

The course includes the following activities and assessments (summary here, details below):

1. **Preparing for and Attending the Class:** We expect you to come to class. Keep on top of the material and if there is something you don't understand, ask about it!
2. **Weekly problem sets:** Learning data analysis takes practice. The problem sets are described below.
3. **In-class midterm:** We will do a pen-and-paper exam on Tuesday, October 13 to help you assess where you are.
4. **Final assessment:** You will complete a cumulative take-home final exam (with computer-use) followed by a short oral exam.

#### 3.1 Preparing for the Class and Precept

Lectures and precept are most effective when you are following along and asking questions. The class is fairly small and will be at its best when we are using the opportunity to discuss the concepts.

Please engage early and often. A key part of being able to engage is reviewing the notes before class so that you know what you do and don't know. Build in the time to do this review!

Reading statistical work (or lecture slides) can be challenging the first time you do it. There will be a temptation to skip over all the math—don't! The math is often where the action is in statistical work. Read carefully and go line by line seriously engaging with material: take notes, write down your impressions or confusions, talk with your classmates, and post questions on Ed (see below). All of your classes should be pushing your research forward and you will be better prepared to integrate this material into your work if you push for a deep level of understanding.

### 3.2 Problem Sets

Statistical methods are tools and it isn't very instructive to read a lot about hammers or watch someone else wield a hammer. You need to get your hands on a hammer or two. Thus, in this course, you will have problem sets on a weekly basis. The assignments will be a mix of analytic problems, computer simulations, and data analysis.

**Format** Assignments should be completed in Quarto which allows you to show both your answers and the code you used to arrive at them. If you haven't seen Quarto before, have a look at the Quarto website and materials on <http://pusocmethodscamp.org/>. We will use Gradescope for submitting and returning the assignments. Your preceptor will provide you with more detailed instructions before the first assignment is due.

**Assessment** The points available for each assignment and question will be stated on the problem set. We value presentable graphs, clear code, readable formatting, and well-written answers. Solutions will be available directly after the problem-set deadlines. Keeping up includes looking at the solutions key to make sure you understand the material! Problem sets will typically be due every week but we will allow you space to prepare for the midterm and final.

**Extensions and Late Policy** You can have *one* no questions asked extension of one week on a problem set of your choosing. If you don't take an extension, we will drop your lowest grade (of any partially completed/submitted problem set). If all your problem sets are completed and with top-level grades (such that dropping the lowest wouldn't help you), we will add a comparable grade bonus to your final assessment. Because we do not want to hold up the class we will not wait for everyone to submit their problem sets in order to post the solutions key. If you are turning your problem in late (with an initial extension or not) you are on your honor to *not look at the solutions* before submitting your work and you are required to explicitly write on your assignment that you have not looked at the solutions. If you are late without an extension (or exceed the one-week extension period given that you had an extension) your grade will drop on the problem set by 10 percentage points of the total available points per day, with a maximum deduction of 70 percentage points for turning it in late. You have until the beginning of reading period (December 8) to submit a late problem set for partial credit but please don't wait that long—the material is very cumulative!

**Extreme Circumstances** I realize that the above extension policy may not be enough to deal with all situations. For example, if a student has a major illness this might disrupt weeks of school work. If an extreme disruption occurs, you need to reach out to me as quickly as possible. We will

assess your situation and then in collaboration with your Director of Graduate Studies we will come up with a plan to get you up to speed as quickly as possible. If you do not reach out to me, then we cannot work with the Director of Graduate Studies to develop a plan.

**Collaboration Policy:** Unless otherwise stated, we encourage students to work together on the assignments, but you should write your own solutions (this includes code). That is, *no copy-and-paste* from other people's code. You would not copy-and-paste from someone's paper, and you should treat code the same way. However, we strongly suggest that you make a solo effort at all the problems before consulting others. It is easy to trick yourself into thinking you understand how to do something when you have someone else prompting you.

### 3.3 Midterm

An in-class pen-and-paper midterm will be held on **Tuesday, October 13**. We will announce its scope, format, and permitted materials in advance.

### 3.4 Final Assessment

The final assessment of the class is a take-home exam followed by an oral exam. The exam is "open-book" in the sense that you can use the slides, your notes, books, and internet resources to answer the questions. However, the final exam must be completed *by yourself*. The exam will be available during the entire period allocated for take-home exams by the university. It will be approximately the length of a long problem set (although we caution that it might take longer if you are used to collaborating on the problem sets). We encourage you to start early.

We will be in touch by the midterm to discuss the timing of the final oral exam.

### 3.5 Grading

The Registrar lists SOC 500 as P/D/F. Non-Sociology graduate students who wish to receive a letter grade should consult the enrollment-change instructions in the [official course listing](#).

Final grades will combine problem sets (55%), class participation (5%), the in-class midterm (10%), and the final assessment (30%).

### 3.6 AI Tools

The biggest change to statistics pedagogy in the 16 years I have been teaching this material is the creation of generative AI tools that are good at math and coding. These tools are clearly helpful in research. It is not yet clear, however, the best way to integrate them into learning, if at all.

In this class, we will try to train you to use these tools responsibly. We would ask that (1) you do not directly put the entire assignment question into a model, and (2) you tell us how you used them (this isn't relevant to our assessment, we just want to learn how people are using them). Throughout the semester we will do some explicit training of how to use these tools together.

A few things to keep in mind:

- This class is designed to prepare you to conduct and evaluate research. At this point, it is not yet possible to put real research problems into generative AI and get reliable high-quality answers. Therefore, generative AI will not be able to do your research for you. That said, it

is possible to take a research problem and break it into smaller parts and then use generative AI as a tool to help with some of those parts. Therefore, this is the skill we will allow you to explore in this class.

- The assignments you will do in this class are a means to an end, not an end in themselves. They are designed to push you in ways that enable you to build new skills. Doing the assignment without the effort is not likely to build skills and will therefore have little value to you.
- Generative AI is not perfect; far from it. You should expect that what you get from generative AI will be wrong occasionally. Programming is a setting where this might be OK, as described more in [this post](#) by Arvind Narayanan and Sayash Kapoor.

The guiding educational aim remains the same: you should understand, check, and be able to explain the work you submit. Assignments are designed to build skills through practice. Completing an assignment without engaging with its reasoning will not accomplish that purpose.

## 4 How to Learn in this Course

If you find this course challenging, you are not alone. Statistics can be challenging and we cover a lot of ground. However, I am confident that you can handle it. In this section of the syllabus I'm going to provide details on some of the forms of support that we offer in this class and pull back the curtain a bit on the pedagogical design.

Your primary responsibilities in this class are to *work hard* and *communicate* with us about what you need. You can't learn if you aren't putting in the time. We can't help if we don't know there is a problem.

The course is designed to provide every tool we can think of to help you learn the material. If you are willing to put in the time, we want to ensure that time is used as effectively as possible.

### 4.1 Resources for Getting Help

There are a few main sources of support in the class.

#### 1. Class and Precept

We strongly encourage you to be an active participant in class and precept. Ask questions during class if you don't understand something that is happening. You don't even have to have a specific question: just raise your hand and let me know that you aren't following what is happening. I'm always happy to stop and go back.

#### 2. Daily Feedback

After every class I will pass around a note card and ask you to write down something about class. You can write something you liked or didn't like. Something you want to understand better or want to hear more about it. Maybe you want to know how a piece of material connects to the broader goals of the class. You can even just draw a smiley face. I will address the questions either in class or on Ed.

### 3. Readings and Slides

If you are studying alone and you hit something you don't understand, your first instinct should be to study the readings and slides. There is a lot of material in the slides and they are intended to be reviewed multiple times, not just seen once during lecture.

### 4. Ed

Ed is a classroom discussion board where you can post questions about the material. You will not be required to post, but the system is designed to get you help quickly and efficiently from classmates, the preceptors, and me. **Unless the question is of a personal nature or completely specific to you, you should not e-mail teaching staff**; instead, you should post your questions on Ed. The course staff will be monitoring the page, but we encourage you to help your classmates as well. A big part of why we use Ed is because reading other people's questions can be really helpful for bolstering your understanding of the material.

### 5. Preceptor Office Hours

Preceptor office hours will be posted on Canvas. You don't need to have a specific question and can just drop in. In past years students have used this as a convenient opportunity to gather to work collectively on the problem sets with someone to support you nearby.

### 6. Instructor Office Hours

I will have a dedicated office hour each week when you can drop by with questions about the conceptual material in the course.

### 7. Problem Set Solutions

As soon as the problem sets are due the key is posted. I know it is really tempting to just turn your focus to the next problem set, but I highly encourage you to check the solutions each week. The class is extremely cumulative and it will help you out to lock down core concepts. Even if a concept in (for example) week 3 seems unrelated to week 4, it will definitely come back in a future week (or I would have cut it!).

### 8. Assessment Preparation

We will provide preparation materials and announce review opportunities before the midterm and final assessment.

### 9. External Consulting Services

Princeton offers numerous statistical consulting services. For example, Q-APS offers statistical and formal theory consulting (<https://q-aps.princeton.edu>). DSS also has consulting and in fact one of our previous preceptors Sofia Avila will be working there. These are not really for problem set help, but can be useful if you just need help understanding a broader concept or building up a specific skill.

This is a lot of resources but if you can think of something else that would be useful to you, we encourage you to come talk to us. Again, if you are willing to put in the time, we can get you a form of support that matches your needs.

## 4.2 How is the Course Designed?

At a high level, this course builds linear regression and causal inference from the basics of probability. September develops probability, random variables, sampling and inference, and causal frameworks.

We then study adjustment for measured confounding and conditional expectation functions before the October 13 midterm. After fall recess, we develop simple and multiple regression, diagnostics and specification, causal regression and sensitivity analysis, and natural experiments and panel designs.

Each week covers a specific topic with a series of closely connected lectures included in a common slide deck. These lectures are designed to fill you in on the core statistical ideas animating the week's topics. The lecture won't focus on code, it will focus on the underlying logic and the applications of the ideas to social science research.

Precepts will teach you the programming tools necessary to implement the things shown in lecture. The code shown in precept is very closely tied to what you will need to complete the problem sets.

The problem sets are where I expect the majority of the learning will be solidified. These assignments are challenging and time consuming, but it is only through carefully engaging with the material that you will cement your understanding of it. If at the end of lecture you feel like you don't have a good handle on the material—that's to be expected. If after the problem set has been submitted you still feel uneasy with the material, you should come to office hours and talk to one of us about it.

Finally, there is a strong focus in the class on understanding why things work rather than just applying them. For example, we will often program up our own functions for things R has built in functionality for. Why do we do that? By programming it ourselves or deriving a known result, we force ourselves to really understand the underlying mechanics. This not only improves our understanding of statistical analysis but it also helps learn new things in the future. Our goal isn't just for you to learn this material, it is to prepare you to teach yourself new material in the future.

### 4.3 Advice from Prior Generations of Students

Each year I ask students to provide advice to future generations of students. Here is some advice from prior students responding to "What advice would you give to another student considering taking this course?" I think the advice is great and it may be helpful coming from other students.

- Be ready to spend a lot of time
- Ask questions if you don't know what's going on!
- Study hard, work hard, review the slides.
- Investing a considerable amount of time in getting familiar with R and its various tools will pay off in the long run!
- Go over the lecture slides each week. This can be hard when you feel like you're treading water and just staying afloat, but I wish I had done this regularly.
- It's challenging but very doable and rewarding if you put the time in. There are plenty of resources to take advantage of for help.
- It will be hard but you will learn so much.
- This course is very challenging but greatly contributed to my understanding of social statistics. If you're truly invested in the subject and willing to put in the work (more than you expect possibly), it will be one of the best courses you've taken.
- This is a course where you will learn a lot and spend most of your time doing the psets. I highly recommend office hours for clarification as lecture covers a lot of material.

## 4.4 A Note to Students Outside the First-Year Sociology PhD Cohort

This course is completely designed to provide training to first year PhD students in the Department of Sociology. That means that some of the topics covered—and the way that these topics are covered—may not be optimal for your particular backgrounds. This warning aside, previous generations of students from many different departments have found the course useful.

## 5 Course Outline

The outline is the working plan for Fall 2026; pacing may change as the semester progresses. Readings are optional references that give another presentation of the material. The lecture slides remain the primary reading. Any required advance reading will be announced in class.

### Lecture 0: Probability Preparation (Tuesday, September 1)

- Special prep session before the first regular class meeting
- Sample spaces, events, and the core ideas of probability
- Marginal, joint, and conditional probability; Bayes' rule; independence
- Introduction to random variables, PMFs, CDFs, and densities

Optional Reading:

- Blitzstein and Hwang, Chapters 1–2 (probability and conditioning), Sections 3.1–3.2 (random variables)
- Aronow and Miller, Chapter 1-2

### Lecture 1: Course Architecture and Core Ideas (Thursday, September 3)

- Course goals, organization, resources, and requirements
- Statistical thinking: the data-generating process and what we learn from data
- Estimands, identification, and estimation
- Learning, programming, and responsible use of AI tools

Optional Reading:

- Lundberg, Ian, Rebecca Johnson, and Brandon M. Stewart. 2021. “What Is Your Estimand? Defining the Target Quantity Connects Statistical Evidence to Theory.” *American Sociological Review* 86(3): 532–565. Opening sections through p. 545; revisited in the causal-inference unit.

### Lecture 2: Random Variables (September 8/10)

- Random variables, distributions, and functions of random variables
- Expectation, LOTUS, variance, and moments
- Joint, marginal, and conditional distributions; conditional expectation
- The laws of iterated expectation and total variance
- Covariance, correlation, and independence; the multivariate Normal distribution

Optional Reading:

- Blitzstein and Hwang, Sections 3.7 (functions of random variables), 4.1–4.2 (expectation), 4.4–4.6 (indicators, LOTUS, variance), 5.1–5.4 (continuous random variables), 6.1–6.2 (moments), 7.1–7.3 (joint distributions), 7.5 (multivariate Normal), and Chapter 9 (conditional expectation)
- Aronow and Miller, Chapter 2

### **Lecture 3: Learning from Random Samples (September 15/17)**

- Populations, samples, estimands, estimators, and estimates
- Properties of estimators: bias, variance, and consistency
- Sampling distributions; laws of large numbers and the central limit theorem
- Interval estimation, hypothesis testing, and power
- Brief extensions: multiplicity and equivalence testing

Optional Reading:

- Aronow and Miller Chapter 3.1–3.2.6 (IID Random variables and estimation), 3.4.1 (confidence intervals) Chapter 3.4.2 (testing)

### **Lecture 4: Frameworks for Causal Inference (September 22/24)**

- Potential outcomes and causal estimands
- The fundamental missing-data problem of causal inference
- Causal identification, directed acyclic graphs, and structural causal models
- Connections between potential outcomes and causal graphs

Optional Reading:

- Lundberg, Ian, Rebecca Johnson, and Brandon M. Stewart. "What is your estimand? Defining the target quantity connects statistical evidence to theory." *American Sociological Review* 86.3 (2021): 532–565. (Up through pg 545 for now)
- Pearl and Mackenzie. *The Book of Why*. 2018.
- Morgan and Winship Chapter 1: Causality and Empirical Research in the Social Sciences
- Hernán and Robins (2020) Chapter 1: A definition of a causal effect
- Morgan and Winship (2015) Chapter 2 (Potential Outcomes), Chapter 3 (Causal Graphs)

### **Lecture 5: Causality with Measured Confounding (September 29/October 1)**

- Simpson's paradox; confounders, mediators, and colliders
- Experiments and the assumption of no unmeasured confounding
- Choosing adjustment variables and the backdoor criterion
- Stratification, standardization, and adjustment estimands

Optional Reading:

- Angrist and Pischke Chapter 2 (The Experimental Ideal)
- Morgan and Winship Chapters 3-4 (Causal Graphs and Conditioning Estimators)
- Hernán and Robins Chapter 3 Observational Studies
- Elwert and Winship (2014) “Endogenous selection bias: The problem of conditioning on a collider variable” *Annual Review of Sociology*
- Aronow and Miller Chapter 7

## **Lecture 6: Estimating Conditional Expectation Functions (October 6/8)**

- Plug-in estimation and nonparametric regression
- Kernels, smoothing, and the bias–variance tradeoff
- Linear prediction and ordinary least squares
- Bootstrap, extrapolation, and transformations

Optional Reading:

- Aronow and Miller, Chapter 4.1.1 (bivariate regression)

## **Midterm and Feedback (October 13/15)**

- **Tuesday, October 13: in-class midterm**
- Thursday, October 15: feedback, corrections, and synthesis.

**Fall Recess: October 17–25. No Class October 20 or 22.**

## **Lecture 7: Simple Linear Regression (October 27/29)**

- Mechanics, assumptions, properties, and inference for ordinary least squares
- Adding a second regressor and the Frisch–Waugh–Lovell theorem
- Indicator variables and interactions

Optional Reading:

- Aronow and Miller 4.1.2 (OLS Regression)

## **Lecture 8: Multiple Linear Regression (November 3/5)**

- Multiple regression in matrix form
- Classical and agnostic inference; sandwich standard errors
- Joint inference, F-tests, and clustered standard errors

Optional Reading:

- Aronow and Miller 4.1.3 (Regression with Matrix Algebra)
- Fox Chapter 9.1-9.4 skip 9.1.1-9.1.2 (Linear Models in Matrix Form)
- Fox Chapter 10 (Geometry of Regression)

## Lecture 9: Regression Diagnostics and Specification (November 10/12)

- Residuals, leverage, influence, and unusual observations
- Functional form, specification error, nonlinearity, and splines
- Measurement error

Optional Reading:

- Fox Chapters 11–13 (regression diagnostics)
- Aronow and Miller Chapters 4.2-4.4 (Inference, Clustering, Nonlinearity)
- Angrist and Pischke Chapter 8 (Nonstandard Standard Error Issues)

## Lecture 10: Causal Inference Using Regression (November 17/19)

- Estimands, heterogeneous effects, and variance-weighted treatment effects
- The parametric g-formula
- Fixed effects and sensitivity to omitted confounding

Optional Reading:

- Cinelli, Carlos, and Chad Hazlett. “Making sense of sensitivity: Extending omitted variable bias.” *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 82.1 (2020): 39-67.

## Lecture 11: Natural Experiments (November 24 and December 1)

- Natural experiments and instrumental variables (November 24)
- Regression discontinuity (December 1)
- Bridge to repeated observations and panel designs

Optional Reading:

- Felton, Chris, and Brandon M. Stewart. 2026. “Handle with Care: A Sociologist’s Guide to Causal Inference with Instrumental Variables.” *Sociological Methods & Research* 55(1): 3–50. [Published article](#); [open preprint](#).
- Angrist and Pischke Chapter 4 Instrumental Variables
- Angrist and Pischke Chapter 6 Regression Discontinuity
- Morgan and Winship Chapter 9 Instrumental Variable Estimators of Causal Effects
- Hernán and Robins Chapter 16 Instrumental Variable Estimation

**Thanksgiving Recess: No Class Thursday, November 26.**

Class meets on Tuesday, November 24.

## Lecture 12: Panel Data and Course Synthesis (Thursday, December 3)

- Difference-in-differences and panel designs
- Cumulative synthesis: targets, identification, estimation, and uncertainty

Optional Reading:

- Angrist and Pischke Chapter 5 Parallel Worlds: Fixed Effects, Differences-in-Differences and Panel Data
- Imai, Kosuke, and In Song Kim. "When should we use unit fixed effects regression models for causal inference with longitudinal data?." *American Journal of Political Science* 63.2 (2019): 467-490.
- Morgan and Winship Chapter 11 Repeated Observations and the Estimation of Causal Effects

## 6 Inspirations

The development of this course has been influenced by a number of people particularly: Matt Blackwell, Dalton Conley, Adam Glynn, Justin Grimmer, Jens Hainmueller, Erin Hartman, Chad Hazlett, Gary King, Kosuke Imai, Kevin Quinn, Matt Salganik, Florencia Torche, and Teppei Yamamoto. I am grateful to everyone who has contributed to these materials, directly or indirectly. I am also grateful to generations of past preceptors who have had a huge influence on the direction the class has gone including Clark Bernier, Emily Cantrell, Elisha Cohen, Max Fineman, Alex Kindel, Angela Li, Ian Lundberg, Shay O'Brien, Alejandro Schugurensky, Ziyao Tian, and Simone Zhang.